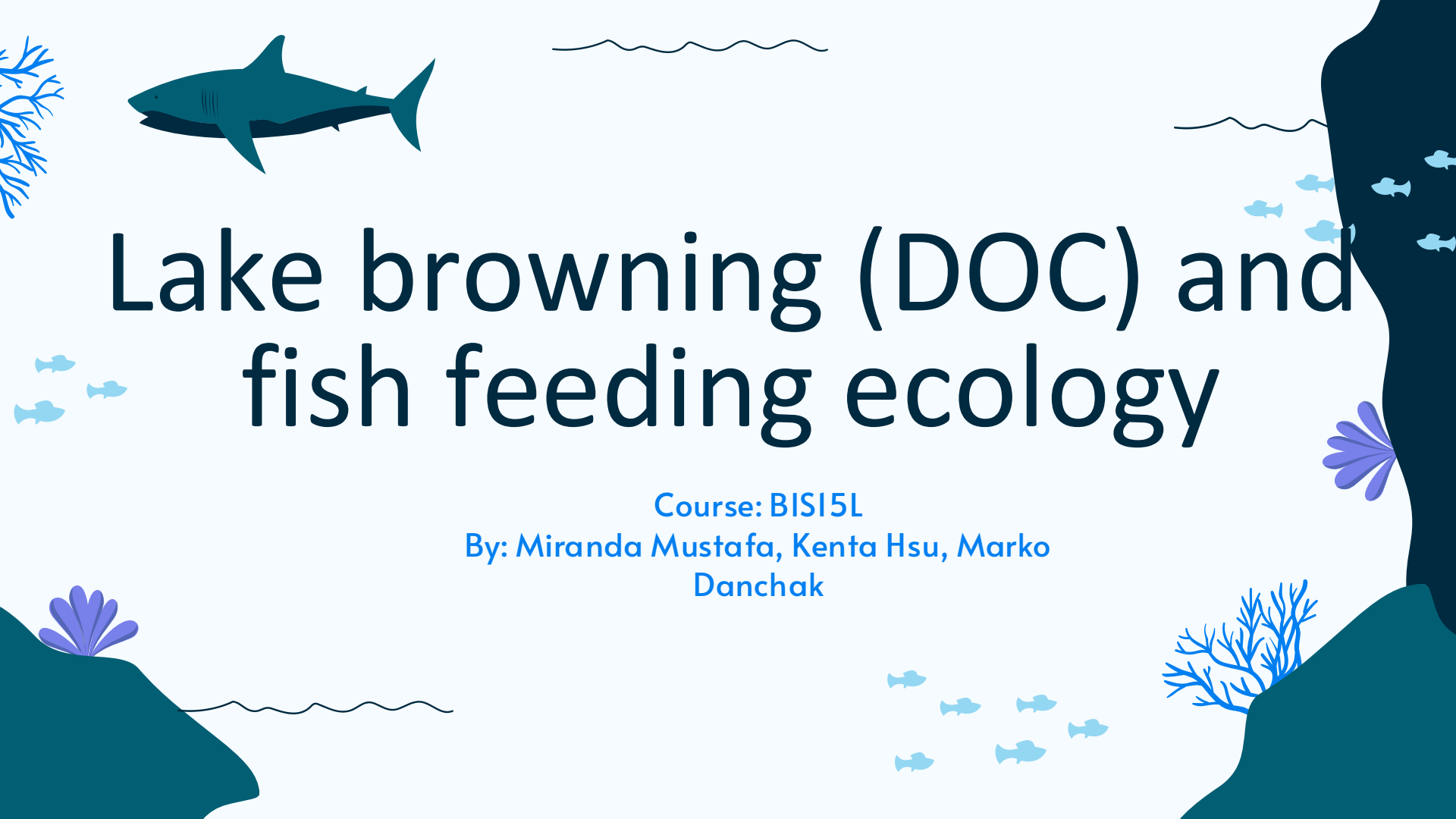


The background is a light blue gradient. In the top left, there is a large, dark teal shark swimming towards the left. To its right, there are two horizontal wavy lines representing the water surface. On the far left, there is a blue coral-like structure. On the right side, there is a large, dark teal rock formation with several small, light blue fish swimming around it. At the bottom left, there is a dark teal rock formation with a purple, fan-like coral. At the bottom right, there is another dark teal rock formation with blue coral and several small, light blue fish swimming around it.

Lake browning (DOC) and fish feeding ecology

Course: BISI5L

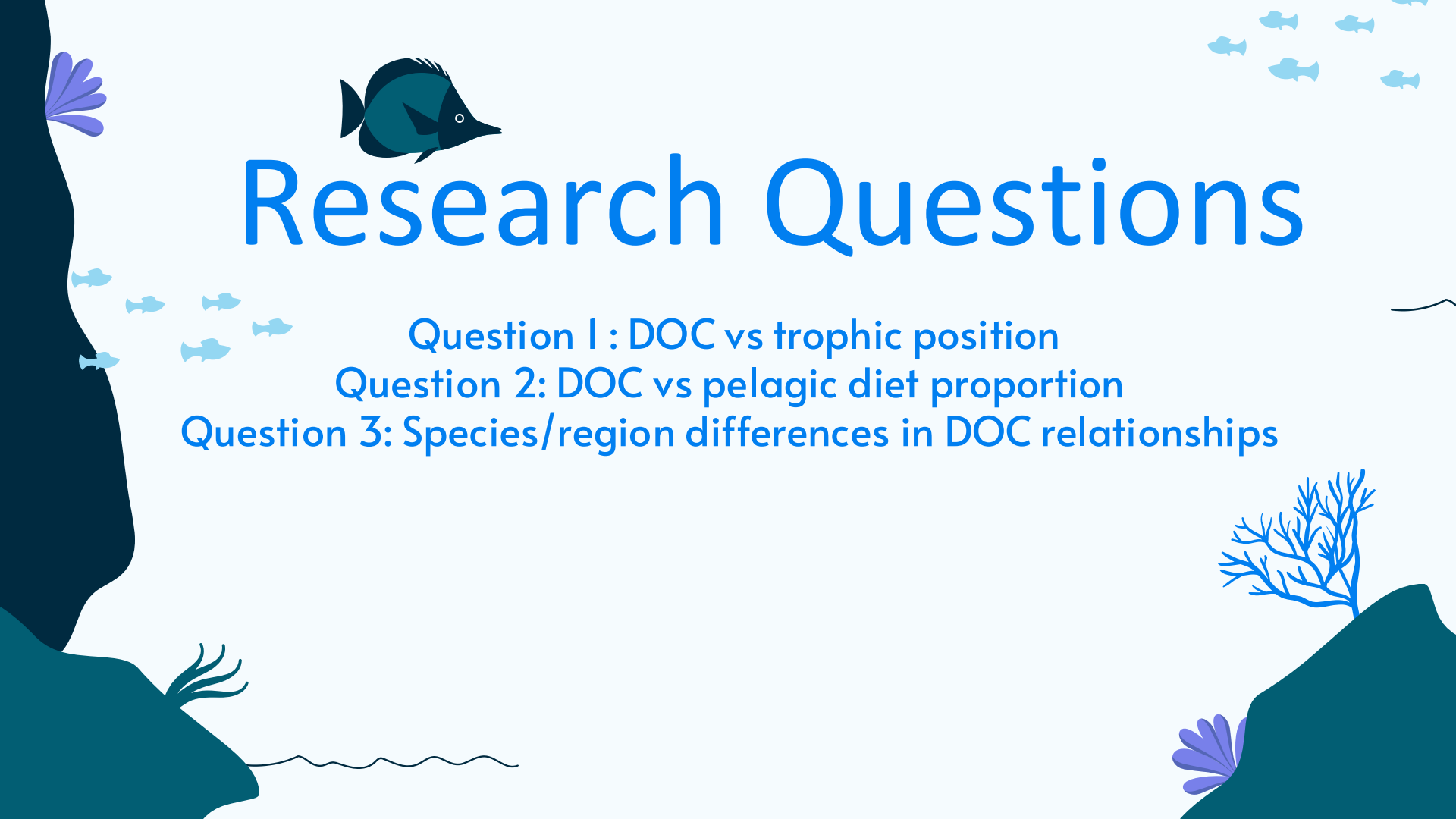
By: Miranda Mustafa, Kenta Hsu, Marko
Danchak

Background

DOC or Dissolved Organic Carbon is when decaying plants or organisms causes the water to brown overtime. This makes it so the sunlight cannot penetrate through the water as easily and thus controls nutrition availability and water temperature. DOC reducing light penetration causes primary production is suppressed and reliance on algae-based food sources shifts to bacteria-based food sources

Data Source

- Charette et al. 2024 freshwater fish dataset - -
3,430 fish records, 78 lakes, 6 species, 6 regions
- Variables: DOC, isotopes, trophic position, pelagic
diet proportion, fish length



Research Questions

Question 1 : DOC vs trophic position

Question 2: DOC vs pelagic diet proportion

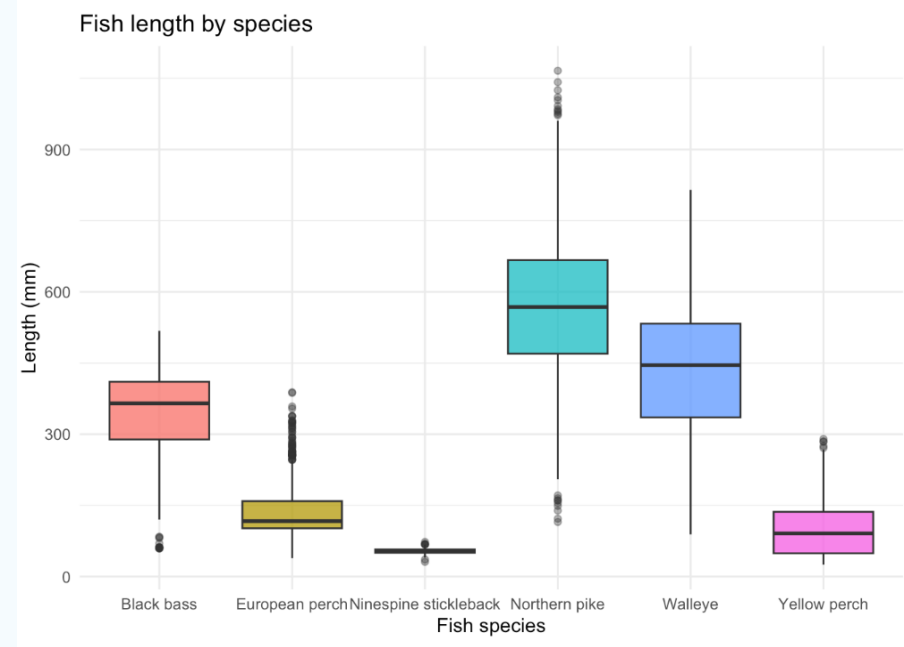
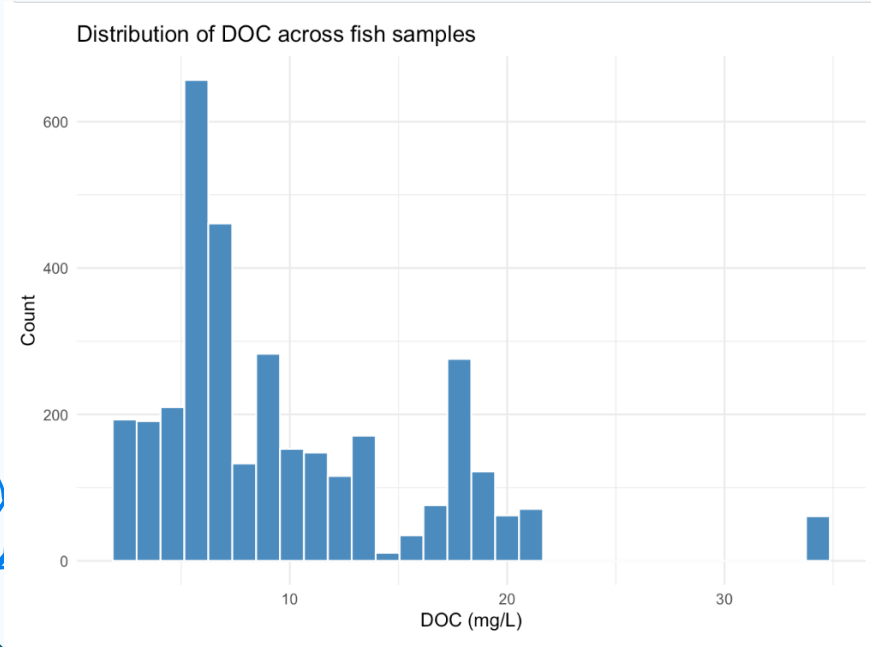
Question 3: Species/region differences in DOC relationships

Workflow

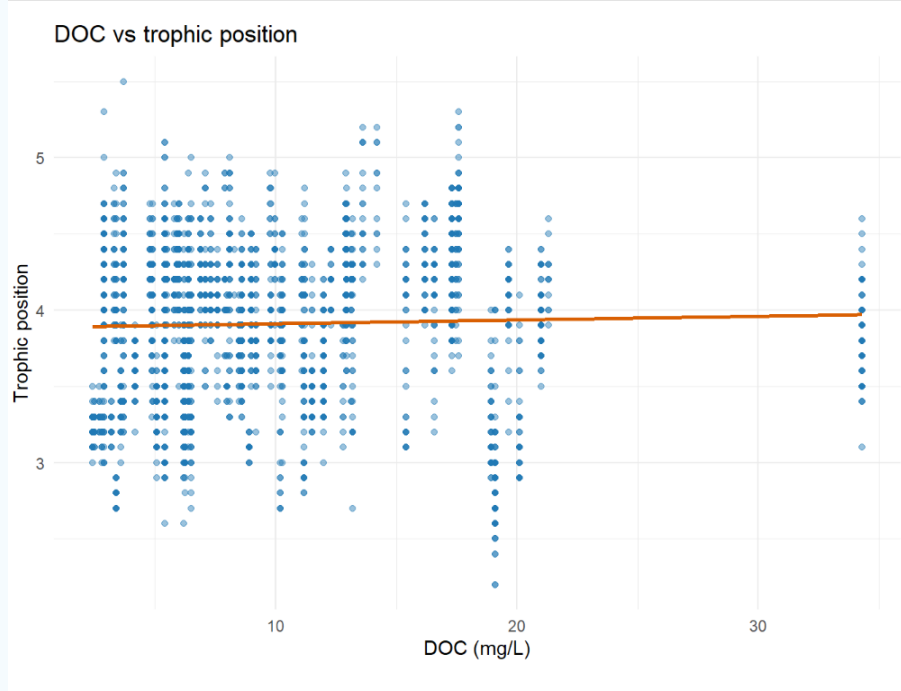
- Import and clean two CSV files
- Join fish and baseline data by lake
- Create analysis dataset and DOC categories
- Check missingness and build summaries



Results



Does DOC relate to trophic position?

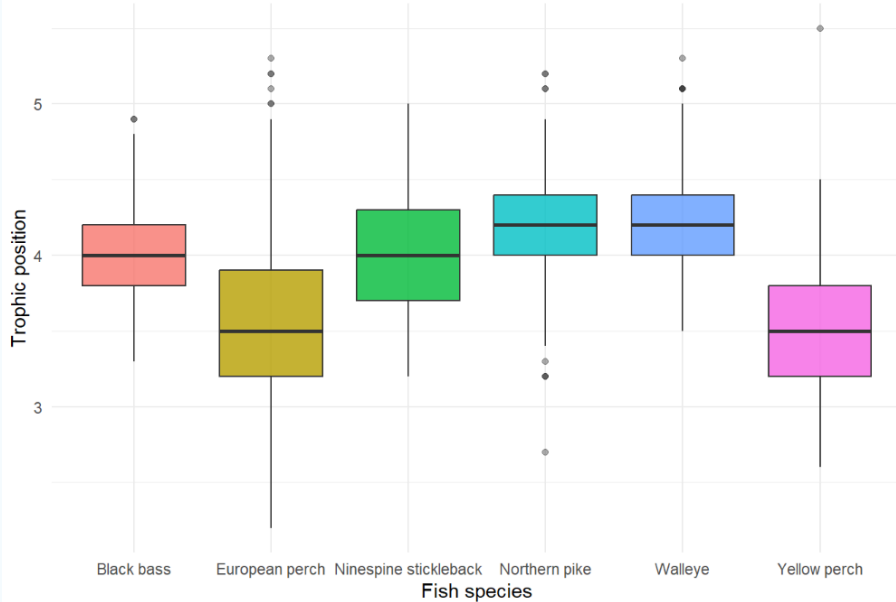


Does DOC relate to pelagic diet

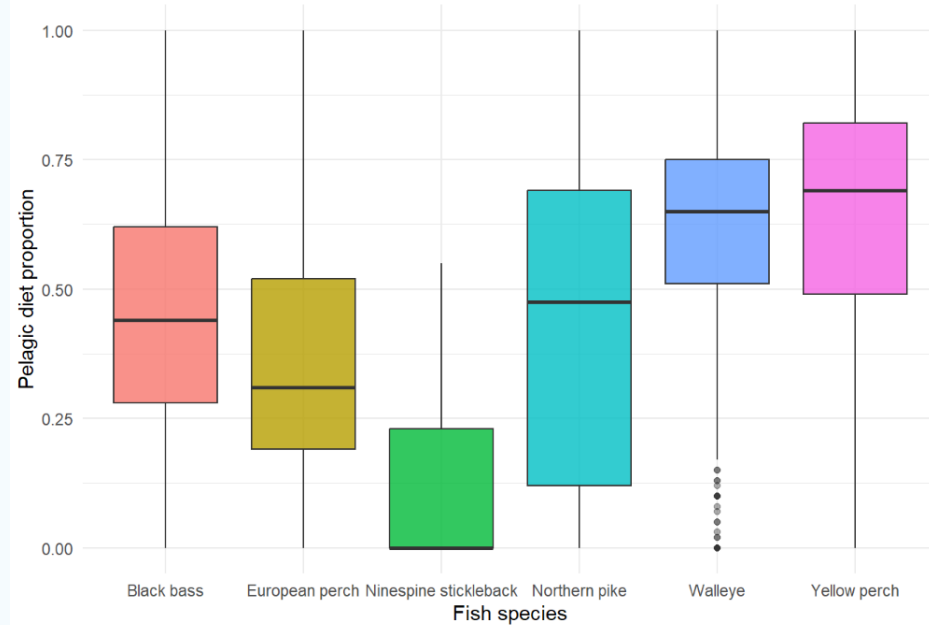


Do DOC patterns vary by species?

Trophic position by fish species

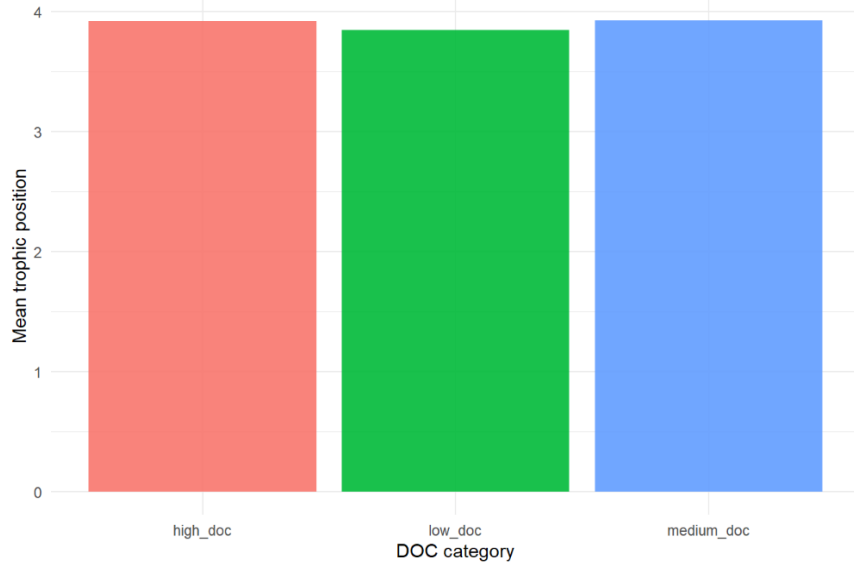


Pelagic diet proportion by fish species

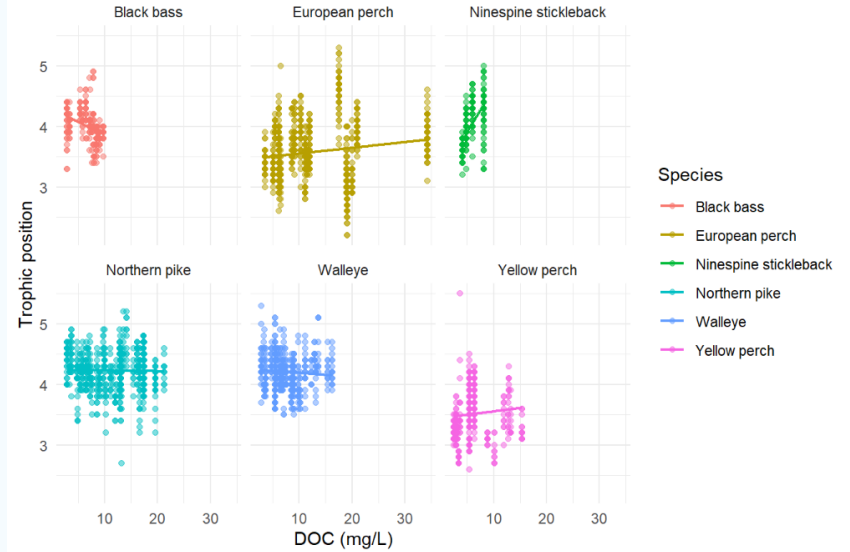


Do DOC patterns vary by species?

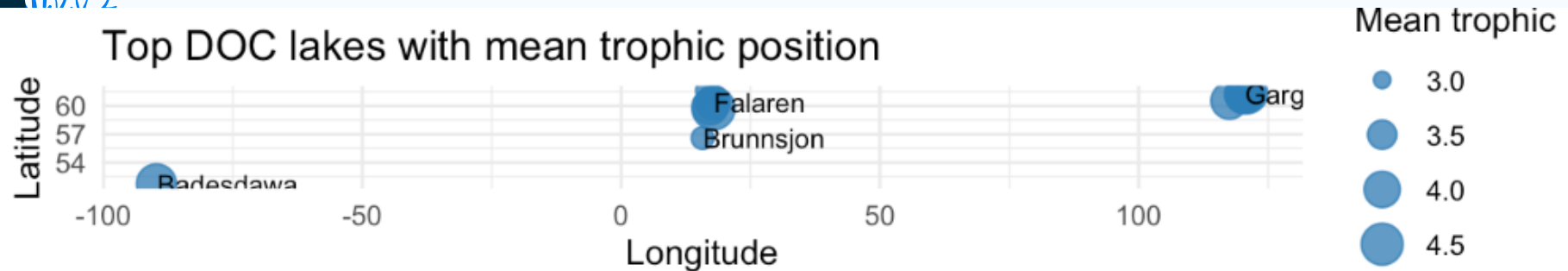
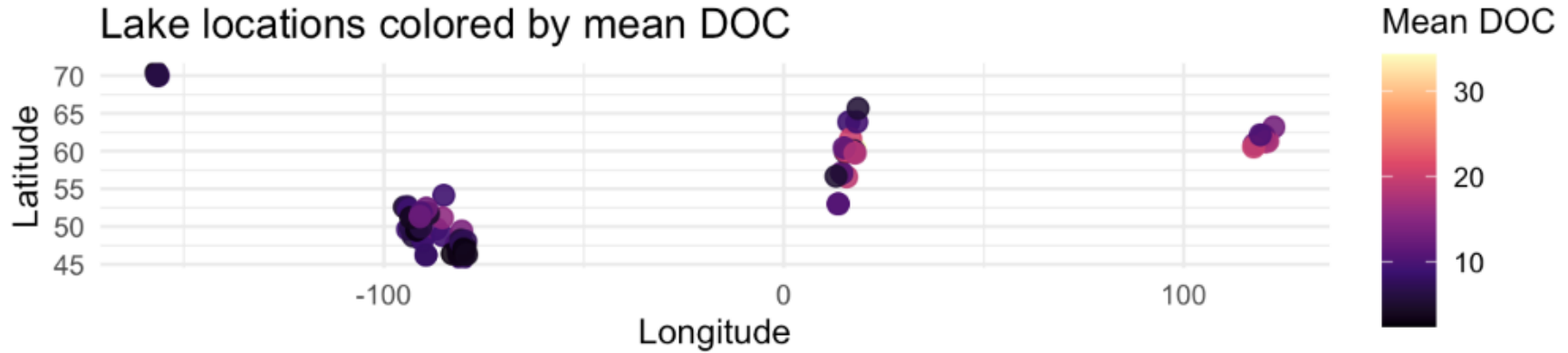
Mean trophic position by DOC category



DOC vs trophic position by species

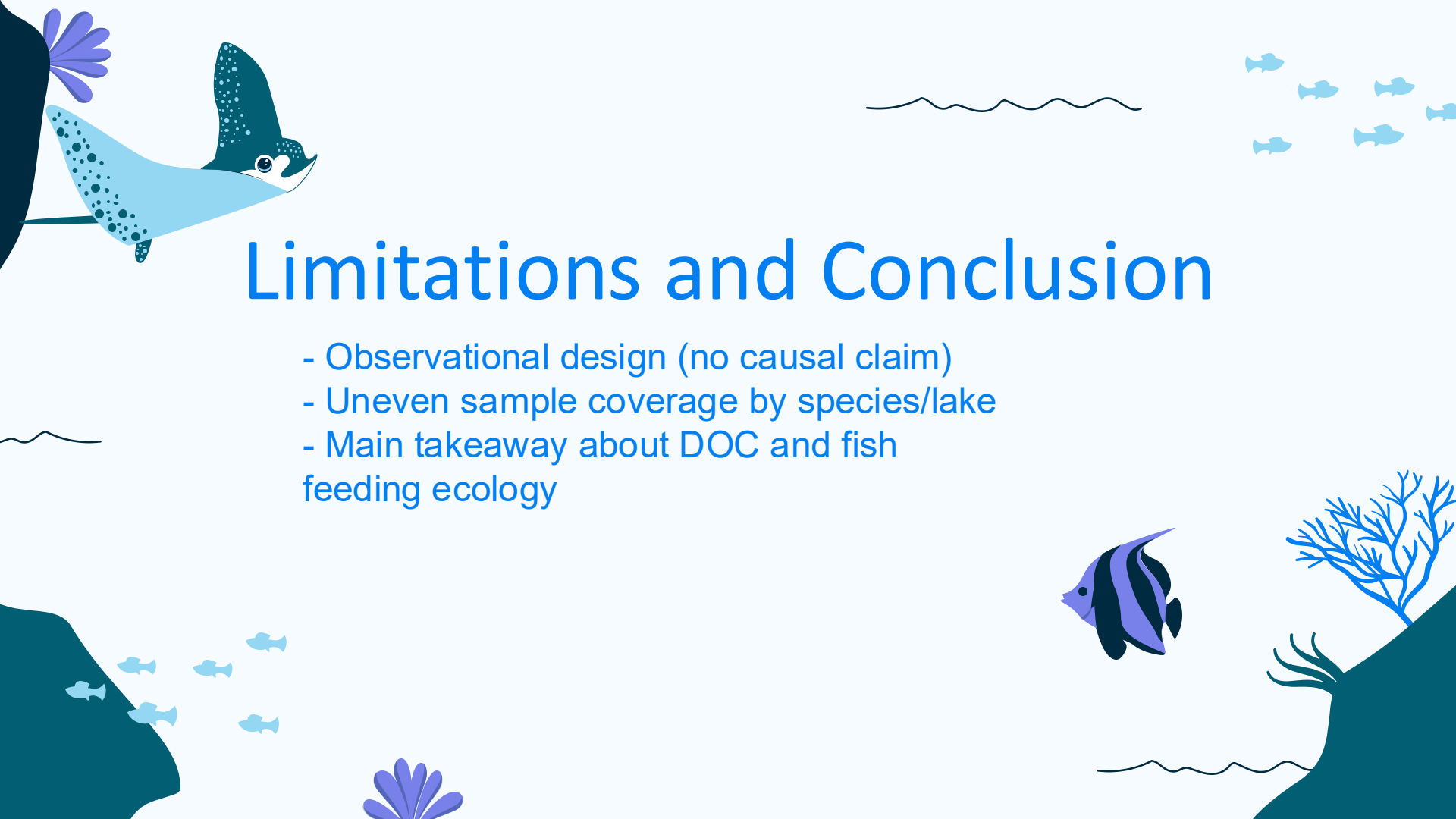


Mapping





Shiny App Demo



Limitations and Conclusion

- Observational design (no causal claim)
- Uneven sample coverage by species/lake
- Main takeaway about DOC and fish feeding ecology